SPEAKER NOTES

Speaker notes: Evaluating eFIx

CS637 video presentation. **Hard limit: 25 minutes** for the whole video.

The notes below are short on purpose. People naturally speak **2–3× longer** than a written script (explaining, pausing, rephrasing), so each section shows both numbers. Use the notes as the points to make, say them in your own words, and stay within your **maximum**. Time yourself once before recording.

Section	Maximum
Ghada Suleiman (slides 1–9)	7:30 min
Suraj Singh Raman (slides 10–17)	7:30 min
Rishikesh Sawant (slides 18–25)	7:30 min
All three (slides 26–27)	1:30 min
Whole video	**24:00 min** (about 1 min margin under 25 for slide changes)

Ghada Suleiman (slides 1–9)

Script read word for word	Natural speech (2–3× the script)	Maximum for your recording
2:41 min	about 5:22–8:03 min	**7:30 min**

Slide 1: Evaluating eFIx

Hello, we are Ghada, Rishikesh and Suraj. We evaluate the ICCPS 2025 paper eFIx: we rebuilt it from scratch, reproduced it on real New York taxi data, critiqued it and extended it.

Slide 2: First the paper, then our work

We first summarise the paper. Then we present our work: evaluation, implementation, reproduction and new ideas. Navy badges mark the paper’s content; orange marks ours.

Slide 3: A taxi fleet as a power plant on wheels

An e-taxi uses over 50 kWh a day, so a Manhattan fleet charges at about 100 megawatts. In emergencies, grids pay big consumers to cut demand within minutes: demand response. Unlike a factory, a fleet can shift when and where it charges without passengers noticing.

Slide 4: A DR program has three stages

A DR program has three stages, Figure 1. In provision, the fleet keeps serving and gets ready, without charging extra. In service, it cuts charging for two hours. In recovery, it recharges. Since the call can come any time, the fleet must always be ready.

Slide 5: Reacting helps; preparing helps more

The paper's motivating example, Figure 2. A fleet that only reacts cuts 127.86 megawatt-hours; one that prepared in advance cuts 139.37, 9% more, serving the same passengers. The paper asks how to prepare best.

Slide 6: Be ready for a call that can come any time

The question: how should the fleet dispatch and charge now, so that whenever a call comes it can cut as much charging as possible without losing passengers? Flexibility is charging with no call minus charging when responding. Three rules: the call time is unknown, no extra charging beforehand, and no worse service.

Slide 7: The model: counting taxis, not tracking them

The model counts taxis rather than tracking them: vacant, occupied, charging or waiting, per region and battery level, in 20-minute slots. Each slot decides where taxis go and how many charge. Everything is linear, so each plan is a linear program.

Slide 8: Why it is hard: the yardstick moves

The hard part: flexibility is measured against a baseline that is itself the policy being optimised, an infinite loop. The paper fixes the baseline to normal service, giving the bi-level problem in Figure 5, but solved exactly it has 1.5 million variables. So it predicts normal service instead, merging both levels into one linear program.

Slide 9: eFlx: a closed-loop fleet controller

eFlx sits between the grid and the fleet, Figure 4. Algorithm 1 solves the linear program, re-predicts normal service and repeats until the plan settles. Algorithm 2 runs this every 20 minutes, always assuming the call may come next slot. Suraj now evaluates it.

Suraj Singh Raman (slides 10–17)

Script read word for word	Natural speech (2–3× the script)	Maximum for your recording
2:29 min	about 4:57–7:26 min	**7:30 min (including the ~1 min demo clip)**

Slide 10: What the paper claims

First, what the paper claims. On 13,000 Manhattan taxis, eFlx charges 20% less during the DR window than normal operation, called TRC, offers 5.5% more flexibility, and comes within half a percent of an Oracle that knows the start time, at about 2% overhead.

Slide 11: Our critical reading: strengths and gaps

Its strengths: an important new problem, an elegant escape from the infinite loop, real-time speed and fair comparisons. But we found gaps, and acted on most. Flexibility is measured against each policy's own baseline, so it can be gamed. eFlx always assumes DR next slot; our P-eFlx fixes that. The service compromise is never measured; we measure it. Only load cuts are studied; we test absorbing power. Parameters are missing, energy use is unrealistic, and there are notation slips.

Slide 12: Rebuilding eFlx with no code released

With no code released, the paper became our specification: model.py holds Equations 1 to 4, policies.py the strategies including eFlx, sim.py the control loop and flexibility score, and fastplan.py the same programs 20 times faster for Manhattan. Every value the paper omits is documented in NOTES.md.

Slide 13: How we know our code is right

To check correctness, the simulator verifies that no taxi is ever created or lost, and two independent builders of the same linear programs reach identical optimal values on four test cases.

Slide 14: Toy model: the core idea reproduces

On a toy city of 2 regions and 100 taxis, the core idea reproduces: eFlx cuts charging during DR by 38% at full service, close to the Oracle. Note the jump back to full charging at 19:20.

Slide 15: Same optimum, different answers

A reproducibility finding: our two builders reach the same optimum but pick different tied solutions, which changes results by up to 14%. So a gap like 0.52% from the Oracle can sit inside solver noise. Next, a quick demo.

Slide 16: Demo: anyone can rerun it

over the screen recording Here is our repository. The validation test takes four seconds and the toy experiment seven, regenerating this figure. The Manhattan runs take hours, so we show saved results.

Slide 17: Real data: NYC taxi trips, March 2024

For real data we used New York's March 2024 taxi records: two million weekday Manhattan trips, grouped into 12 regions since the paper's 38 aren't public. We kept the paper's battery and timing values. Rishikesh now presents the results.

Rishikesh Sawant (slides 18–25)

Script read word for word	Natural speech (2–3× the script)	Maximum for your recording
2:30 min	about 5:00–7:30 min	**7:30 min**

Slide 18: Figs. 6 and 7: paper vs ours (3,200 taxis)

The paper's Figures 6 and 7 beside ours. At the paper's own 13,000 taxis, every policy cuts DR charging to zero, so nothing can be compared; we use 3,200 taxis. There the ordering holds in direction, with eFlx charging least, then R2I, then normal operation, but the error bars are large.

Slide 19: Claim by claim: direction holds, size doesn't

Claim by claim: eFlx cuts DR charging by 4.8%, not 20%, and lands 4.6% from the Oracle, not half a percent. Flexibility, plus 4% against 5.5%, is the closest match. Without the charging limit it is even worse than R2I and TES, and nearly all the gain comes from one day.

Slide 20: Overheads, and what the experiments miss

Overheads are higher than claimed: 10% more idle driving and 44% more waiting, against 2%. After 19:20 charging spikes to 26 megawatts, a new peak the paper never measures. And at 13,000 taxis eFlx scores 40% more flexibility while every policy charges zero: the metric problem.

Slide 21: Fig. 11 and Table 1: lead time and forecast error

Figure 11: longer preparation helps on the toy city, but on Manhattan only partly, peaking at 80 minutes. Table 1: with a 10% forecast error, flexibility drops 6.8%, not 1.3%, about 5% of passengers are missed, and eFlx's edge almost vanishes.

Slide 22: New use cases: when does eFlx help?

A fleet-size sweep: from 13,000 taxis down to 4,000, any policy can cut DR charging to zero. Only at 3,200 does some charging remain. So the paper's fleet sits in the easy corner of its own problem. We also varied the DR start time.

Slide 23: Absorbing surplus solar and wind power

Grids with surplus solar and wind want fleets to charge more. Reversing the response, our fleet can absorb only 0.1 megawatt-hours extra, because normal operation already uses every free charger. Offering both directions needs a two-sided plan.

Slide 24: P-eFlx: plan for when DR is likely

Our first idea, P-eFlx, plans with a probability of when DR is likely instead of assuming the next slot. It cuts DR charging by 32% versus eFlx on the toy city and 2.6% on Manhattan, closing three quarters of the gap to the Oracle without knowing the start time.

Slide 25: How much flexibility does a little service buy?

Our second idea prices the service compromise. On Manhattan, missing about 4% of passengers gives two and a half times the flexibility, while eFlx's provisioning is worth under 1% of service.

All three (slides 26–27)

Script read word for word	Natural speech (2–3× the script)	Maximum for your recording
0:33 min	about 1:07–1:40 min	**1:30 min**

Slide 26: What the paper did, and what we did

Ghada: The paper turned a hard bi-level problem into one fast linear program, claiming 20% less charging.

Suraj: We rebuilt and validated it; our limits are chosen parameters, 12 regions and a fluid simulator.

Rishikesh: On real data the idea holds, but the gain shrinks to about 5%, and we added three new ideas.

Slide 27: Contribution

Ghada: Thank you for watching.

Suraj: All our code, data and results are in our GitHub repository.

Rishikesh: We look forward to your questions.

Script total: 1155 words, about 8.2 minutes if read word for word.